



Voltage, Current, Power, Resistance and Energy

Current: $I = \frac{\Delta Q}{\Delta t}$: the rate of charge flow

Ohm's law

$$V = I R$$

if you increase I , V will increase because: $V = I R$

↖ bigger number here will give bigger V

if you increase R , I will decrease: $I = \frac{V}{R}$

↖ bigger number here will give smaller I

Power

$$P = IV$$

$$V = IR$$

$$P = I(IR)$$

$$P = I^2 R$$

$$P = IV$$

$$I = \frac{V}{R}$$

$$P = V\left(\frac{V}{R}\right)$$

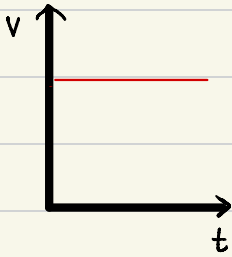
$$P = \frac{V^2}{R}$$

Power: Rate of transfer of energy

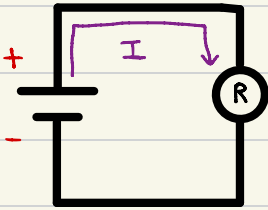
$$P = \frac{E}{t}$$

Energy: $P \times t$

Dc Current and AC Current



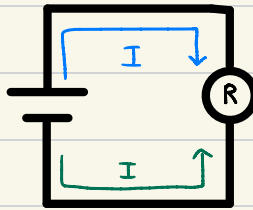
D.C



Current flows in a
single direction



A.C



eg. If it switches 60 x
per second, It will have
a $f = 60 \text{ hz}$ (frequency)

currents direction keeps switching
per second

Peak Voltage and Peak current

V_o = peak voltage

I_o = peak current

$$V_o = \sqrt{2} \times V_{rms}$$

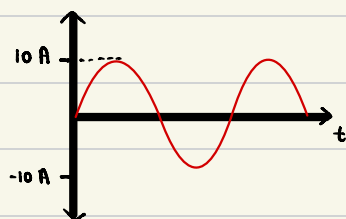
$$I_o = \sqrt{2} \times I_{rms}$$

E.g1) The rms voltage in a house is 120 V , calc. peak voltage

$$V_0 = V_{rms} \times \sqrt{2}$$

$$V_0 = 120 \times \sqrt{2} = 169.7 \text{ V}$$

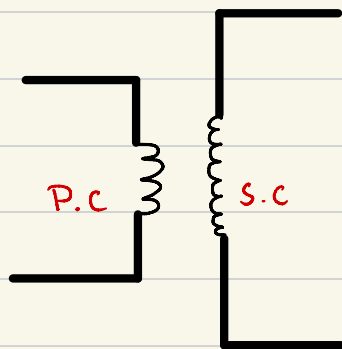
E.g2) From the graph below , calculate the rms current



$$I_0 = I_{rms} \times \sqrt{2}$$

$$\Rightarrow I_{rms} = \frac{I_0}{\sqrt{2}} \Rightarrow \frac{10}{\sqrt{2}} = 7.07 \text{ A}$$

Transformers and Electromagnetic Induction



N = no. coils

V = voltage

I = current

P = power

$\square_p$ = Primary

$\square_s$ = Secondary

$$N_p = 100$$

$$N_s = 1000$$

$$V_p = 12 \text{ V}$$

$$V_s = 120 \text{ V}$$

$$I_p = 10 \text{ A}$$

$$I_s = 1 \text{ A}$$

$N \uparrow \quad V \uparrow \quad I \downarrow$

$$\underline{P_p = 120 \text{ W}}$$

$$\underline{P_s = 120 \text{ W}}$$

$$P_{in} = P_{out}$$

Energy is conserved

Step up transformer

→ secondary coil has more coils than primary

$N \uparrow$ $V \uparrow$ $I \downarrow$ P constant

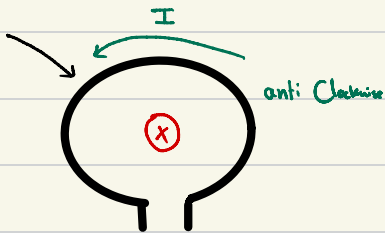
$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_p}{I_s}$$

and

$$P_{in} = P_{out}$$

Electromagnetic Induction

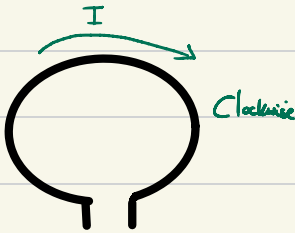
coil



move magnet into the coil, current will be induced



⊗ : magnetic field into page



move magnet away, current will be induced



- moving magnet **slower** will **reduce** the amount of current being induced
- **speeding up** magnet will **increase** induced current
- **stopping** the magnet **won't** allow induce current in the coil at all

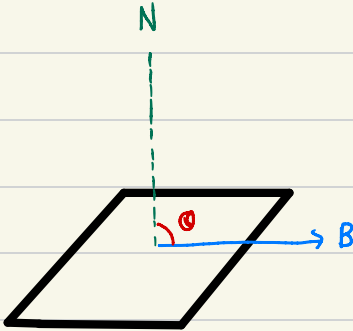
Magnetic Flux

$$I_B = B A \cos \theta$$

I_B = Induced current

WB = magnetic flux

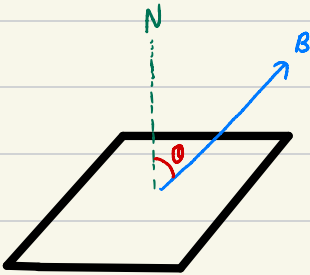
N = normal line



$$\cos(90) = 0$$

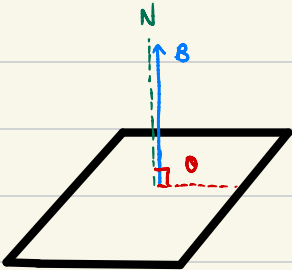
$$\therefore I_B = B \times A \times 0$$

$$I_B = 0$$



$$\cos \theta$$

$$\therefore I_B = B \times A \times \cos \theta$$



$$\cos(0) = 1$$

$$\therefore I_B = B \times A \times (1)$$

$$I_B = B \times A$$

$$\mathcal{E} = -N \frac{\Delta I_{\Phi}}{\Delta t}$$

$\mathcal{E}$ = Induced E.M.F

N = no. coils

ΔI_{Φ} = change in electric flux

$$\mathcal{E} = -N \times \frac{(BA \cos \theta)}{\Delta t}$$

or

Induced current

you can treat $\mathcal{E}$ as Voltage (same unit V)

if $V = I R$ then $\mathcal{E} = I_{\Phi} R$

Periodic table and structure of an atom

Group 1 elements

↳ H, Li, Na, K, Rb $\Rightarrow$ Alkali metals $\Rightarrow$ very reactive

↳ 1 valence electron : After reacting they become '+' ions
 $\Downarrow$
 cations

Group 2 elements

↳ Be, Mg, Ca, Sr $\Rightarrow$ Alkali earth metals

↳ 2 valence electrons : form +2 ion after reacting

Group 8 elements

↳ He, Ne, Ar, Kr, Xe $\Rightarrow$ Noble gases $\rightarrow$ very unreactive

↳ Inert, Already stable, don't react much at all

Group 7 elements

↳ F, Cl, Br, I $\Rightarrow$ Halogens

↳ 7 valence electrons : After reacting the form -1 charges (anion)

Group 6 elements

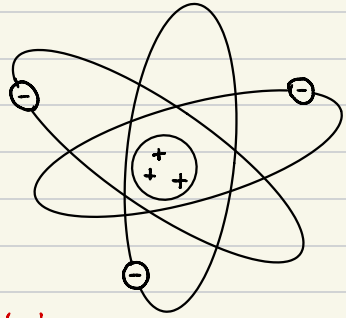
↳ O, S, Se, Te $\Rightarrow$ Chalcogens

↳ 6 valence electrons : For -2 anions

Atomic structure and energy levels :

Rutherford model :

- atom contains nucleus
- protons and neutrons are present in it
- Electrons are orbiting around the nucleus
- The charge on the atom is neutral '+' = '-'



Electron energy levels and Bohr's model

- Valence shell : the outer most shell of an electron.
- Valence electrons : the electrons found in the outer most shell

Numbering shells of an Atom

K , L , M , O , P , Q

← All are in order of
being closest to the
nucleus, Starting with
• K



each shell can only
store a certain number
of electrons.

using the equation $2n^2$, you can find the no. of electrons in
a certain shell.

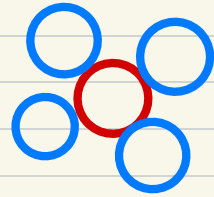
e.g M is the 3rd shell ,

$$\text{therefore ; } 2 \times (3)^2 = 18 \text{ (max electrons)}$$

Timeline of Atomic models

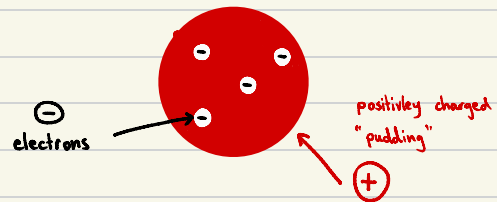
Dalton's Model (1808):

- matter consists of indivisible atoms
- atoms are arranged in different ways to make different compounds



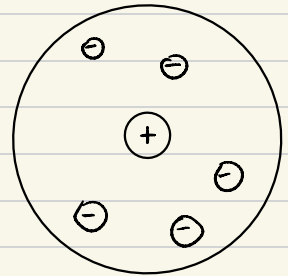
Plum Pudding Model $\Rightarrow$ Thompson (1904)

- electrons are much smaller than atom
- Atom is a positively charged object with tiny electrons put in it



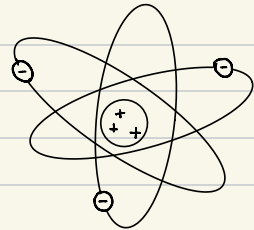
Rutherford model $\Rightarrow$ (1911)

- atom contains nucleus
 - protons and neutrons are present in it
 - Electrons are orbiting around the nucleus
- The charge on the atom is neutral '+' = '-'
- Atoms are mostly just empty space



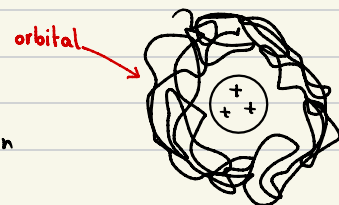
Bohr model: $\Rightarrow$ (1913)

- Electrons orbit the nucleus in a specific path like the planets around the Sun



Quantum Mechanical model: $\Rightarrow$ (1920)

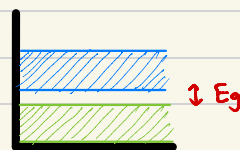
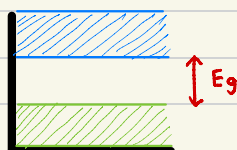
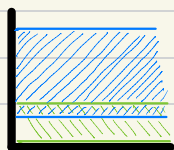
- Electrons weren't moving in orbits, they circled in 'orbitals'. The way the electrons were moving was in a hyperactive manner.



Conductors, Insulators and Semi-Conductors

 : valence band

 : Conduction band



Conductor

In good conductors/metals Energy bands can overlap

When increasing Thermal energy, the electrons will get excited and all the electrons will be free

Insulator

The highest energy band, will be completely filled with electrons

There is a big gap between the 2 energy bands (forbidden energy level). Electrons cannot be in that gap. AKA E_g

If the $E_g > 4 \text{ eV}$, the material is classified as an insulator

Semi-conductor

The highest energy band is filled again. ($E_g < 2 \text{ eV}$: semi-con.)

The energy gap is very small

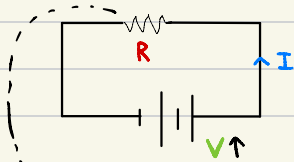
If you increase temp, the electrons will get excited, and due to small E_g , they can easily jump into the conduction band.

The breaking of a covalent bond is equivalent :

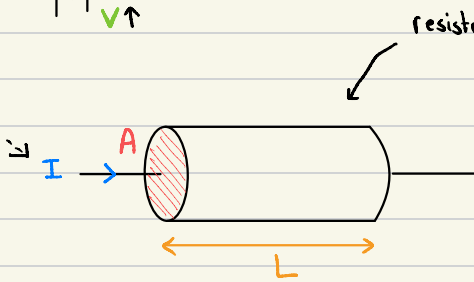
(E required to break covalent bond is same as forbidden energy)

- raising an electron from valence band to conduction band
- moving an electron from an atom, making it a free electron
- of leading to the formation of a hole.

Resistivity and Conductivity



$$V = IR$$



resistor: The resistance depends on the **Cross-sectional Area**, **Length**, **V** and **I**

- If you increase the **L**, it's reasonable to say that **R** will increase as **I** has to flow through a longer resistor.

The bigger the resistivity, the it presents naturally the flow of **I**.

- If you increase **A**, make it wider, it makes sense to say that **I** has more space to go through, so **R** decreases.

$$\rho = \frac{R \times A}{L} \quad \Omega m$$

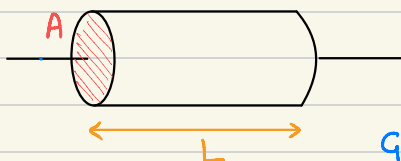
Conductivity

$$\rho = \frac{1}{\sigma} \quad \text{Conductivity}$$

Conductivity: is the ability to allow electric current to flow

The conductance G is the ability of molecule to allow the electric current flow

$$G = \frac{\sigma \cdot A}{L}$$



G : Conductance

σ : conductivity

A : Area

L : Length

$$\frac{1}{\text{ohm}} = \text{mho}$$

Intrinsic and extrinsic Semi-conductors

Intrinsic : • pure materials

- conductivity based on thermal energy exciting electrons into conduction band.

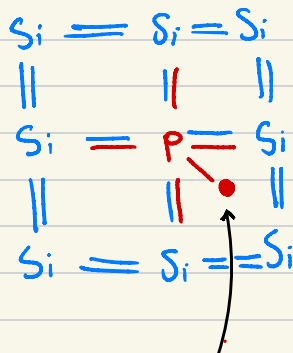
(Si, GaAs) silicon and gallium arsenide

Extrinsic: • doped material (adding small amounts of impurity)

- small amount of a pentavalent (Antimony, phosphorus or Arsenic) are added to Si or Ge.

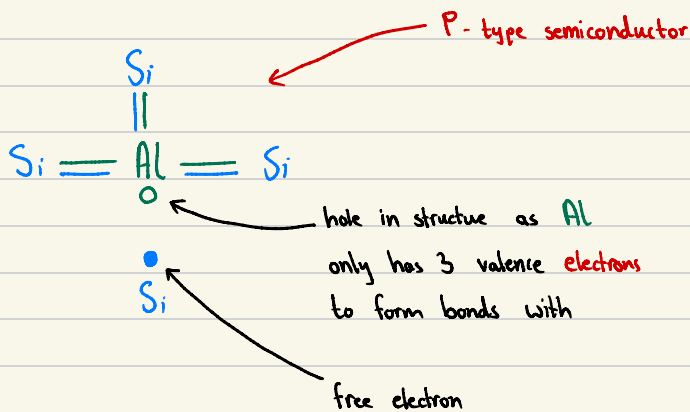
• four of its valence electrons form a pair with Ge (or Si) while the fifth electron gets loosely bonded with right just below the conduction band.

• At room temp, this 5th electron dissociates from it and moves through as a conduction electron



N-type Semiconductor

free electron



In N-type semiconductors, the charge carriers are electrons

In P-type semiconductors, the charge carriers are holes, Electron vacancies.

Conductivity of a Semi-Conductor

$$\sigma = n \times q \times \mu \Rightarrow \text{mobility is very high if the atom is stable. } (\mu \approx 0)$$

ohm cm

no. of electrons

concentration of e per cm^3

charge

1.650×10^{-19}

$\frac{\text{cm}^2}{\text{volt} \cdot \text{sec}}$

Conductivity of Intrinsic semi-conductor :

$$\sigma = n_e \times q \times \mu_e + n_p \times q \times \mu_p$$

conductivity of elect.

conductivity of the holes